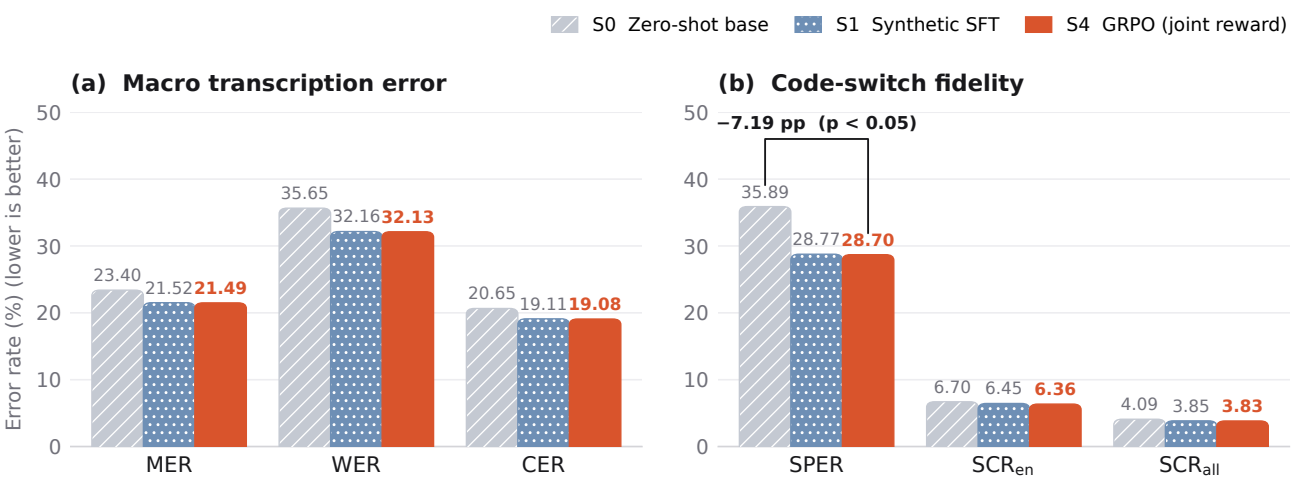

S0 Zero-shot base
S1 Synthetic SFT
S4 GRPO (joint reward)
(a) Macro transcription error
Error rate (%) (lower is better)
50
40
30
20
10
0
23.40
21.52
21.49
35.65
32.16
32.13
20.65
19.11
19.08
MER
WER
CER
(b) Code-switch fidelity
−7.19 pp (p < 0.05)
35.89
28.77
28.70
6.70
6.45
6.36
4.09
3.85
3.83
SPER
SCR$_{en}$
SCR$_{all}$